



Dry lab notebook

Genetic switch

I. General research

[1/08/2023 - 11/09/2023]

Purpose of the genetic switch: ability to toggle between biosurfactant expression and campesterol production in order to prevent overworking the cell

Since no one from our team had any experience or expertise in genetic switches, we looked for inspiration in the literature and reached out to experts for guidance and advice.

Experts

On 22nd of August we reached out to Alexander Fedorec from UCL (suggested by Vitor, our PI).

Meeting with Alex Fedorec [30/08/2023]:

On the 30th of August Julia from our team met with Alexander Fedorec who is a group leader in the Division of Biosciences at University College London in UK. Alex has an extensive expertise in genetic switches and he agreed to talk to us about it and help us get started with designing a genetic switch on our own.

During the meeting we talked about our struggles with designing our own bioswitch and our lack of knowledge in that field. We explained what we want our switch to do and what are our current general ideas. We also expressed our frustration with the fact that because of the characteristics of our project, we can't really rely on concentrations of the biosurfactant or campesterol which would have been an easier approach for the switch.

Alex suggested that we look into a toggle switch since it seems like it would be perfect for our purpose and relatively easy to design at the same time. He also sent us an article¹ that could be useful and told us about a Rodrigo Ledesma-Amaro's Lab – a research group that is developing and looking into synthetic biology tools in *Yarrowia lipolytica*.

II. Brainstorming on the ideas

[24/08/2023 - 14/09/2023]

Problem: Biosurfactant is secreted so detecting the concentration would be very hard. Campesterol is not secreted but it's detection would also be very challenging.

Since we couldn't really use the concentration of either the biosurfactant and campesterol, we had to explore ideas other than a simple feedback loop dependent on the concentration.

We looked into other possible triggers and we've had few brainstorming sessions on it. We looked into a RAD switch² but we decided to go for something a bit simpler due to limited time and lack of broad knowledge of genetic switches. We also thought of a temperature regulated switch but were not sure how could we regulate the temperature and what the details of such a switch could be.

Meeting with Alex and Vitor [12/09/2023]:

On the 12th of September Julia had a chat with Alex and Vitor in the Rega Institute in Leuven. We talked about our troubles regarding designing our genetic switch. We thought of designing a thermal toggle switch³ but were not sure how to implement it and how would the cell change temperature. Vitor mentioned that it would be very interesting since normally during steroid synthesis, the temperature rises and it should be constantly brought down. We could just stop the cooling for a moment, let the temperature rise causing the switch from one to the other state. This approach would also save the energy.

Since we can't really rely on detecting the concentration of biosurfactant or campesterol, we also thought of using metabolic load as trigger for the switch and both Alex and Vitor said that it could also be a good idea.

We also talked about the possibility of exploring the morphological transition from yeast-to-hyphal growth in *Yarrowia* as the trigger for the switch. This could act as a "one time" switch where in the yeast growth phase, the cell would produce biosurfactant and in the hyphae form it would switch to campesterol production.

The idea of implementing a quorum sensing system in the switch was also proposed but we need to look into availability of these systems in our organism.

In addition, Alex advised us to not be shy with the models and try out as much as possible. He suggested that instead of overfocusing on the details (like we did up until that point), we should have fun with the switch and play with it. All insights and suggestions were very helpful and we stay in touch with Alex for further chats.

References

1. Leon, M., Woods, M. L., Fedorec, A. J. H. & Barnes, C. P. A computational method for the investigation of multistable systems and its application to genetic switches. *BMC Syst Biol* **10**, (2016).
2. Bonnet, J., Subsoontorn, P. & Endy, D. Rewritable digital data storage in live cells via engineered control of recombination directionality. *Proc Natl Acad Sci U S A* **109**, (2012).



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3. Gardner, T. S., Cantor, C. R. & Collins, J. J. Construction of a genetic toggle switch in *Escherichia coli*. *Nature* **403**, (2000).